

4.1 The structure of the Earth and distribution of earthquakes and volcanoes

- 1** Label the following in Figure 4.1: crust, mantle, outer core, inner core.

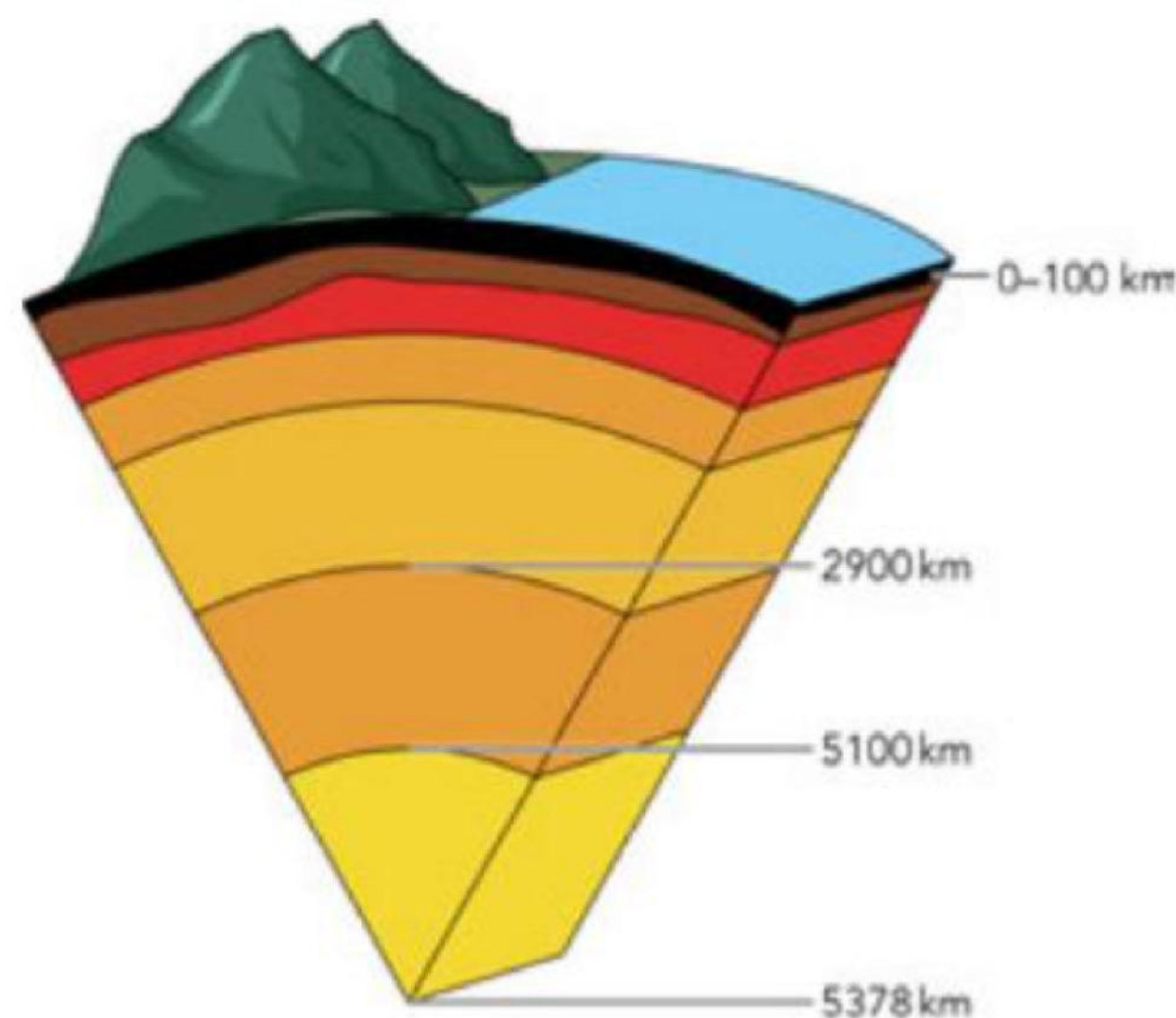


Figure 4.1: The structure of the Earth

- b** What is meant by the term lithosphere?
- c** Copy and complete Table 4.1, giving the state of each layer of Earth: 'solid', 'semi-molten' or 'liquid'.

Earth layer	State
crust	
mantle	
outer core	

Earth layer	State
inner core	

Table 4.1: The states of the layers of the Earth

2 What are the two types of tectonic plates?

26

3 What is the plate boundary type shown in Figure 4.2?

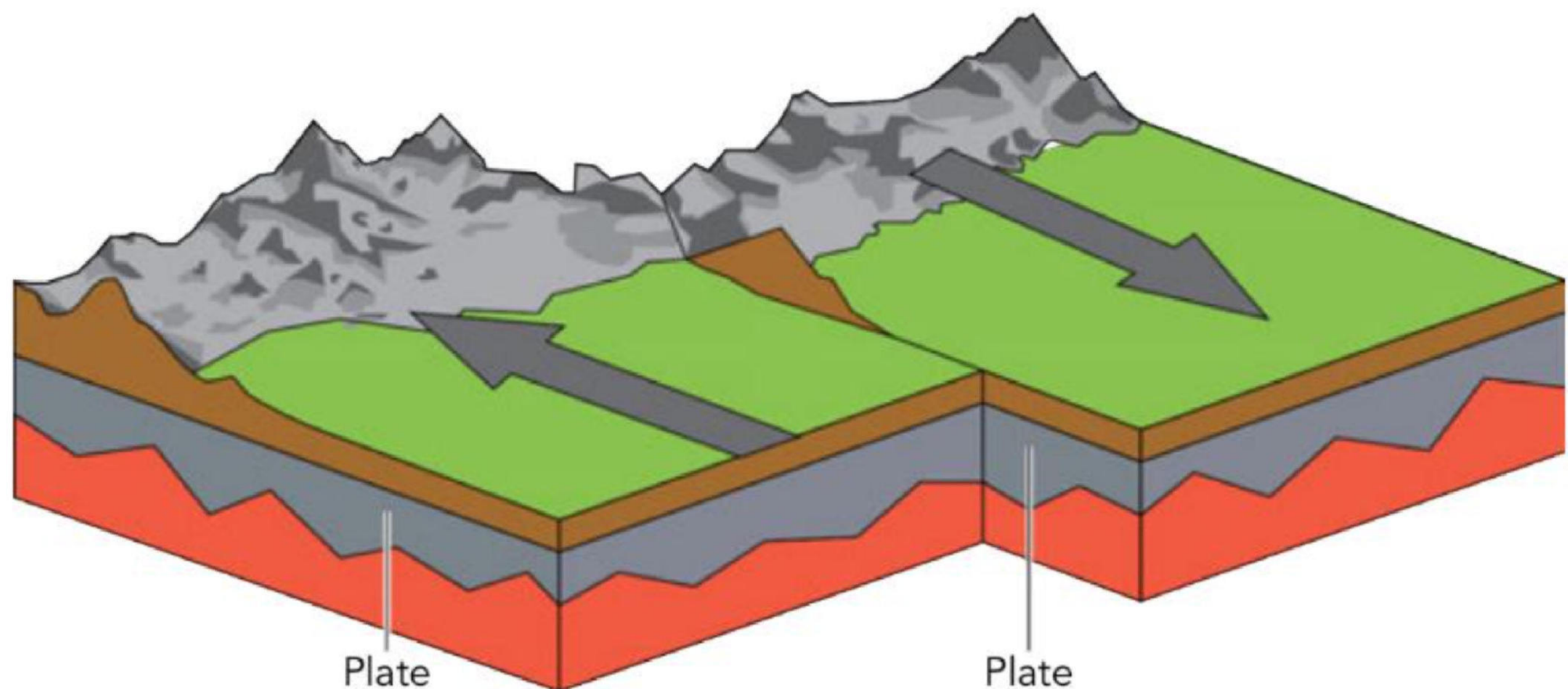


Figure 4.2: A plate boundary

'Identify' questions require you to name/select/recognise something. This could include identifying something from a source. Questions without a source will require you to use your own knowledge, whereas questions that use a source will need you to use it in some way.

4 Identify the plate boundary type shown in Figure 4.3.

[1]

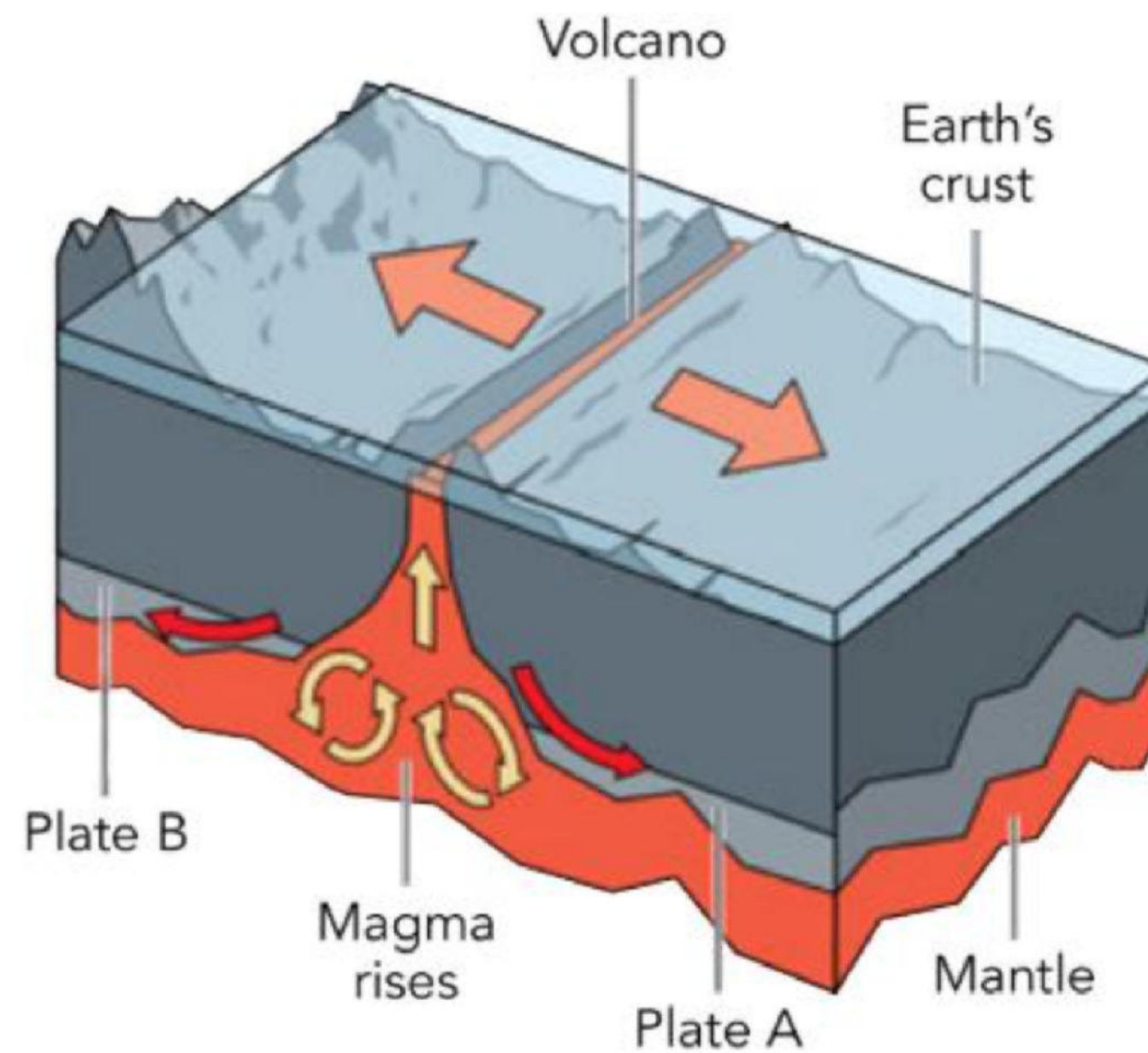


Figure 4.3: A plate boundary

- ~~5~~ Identify **one** tectonic hazard found at a convergent/destructive plate boundary. [1]

Questions that use the command word 'suggest' require you to apply your knowledge – remember that there is not one fixed answer, so you will have to use logical thinking.

- ~~6~~ Suggest **one** reason an oceanic plate subducts beneath a continental plate. [1]

- ~~7~~ Suggest **one** reason why there are no volcanoes at a transform plate boundary. [1]

UNDERSTAND THESE TERMS

- Crust
- Mantle
- Lithosphere
- Core
- Magma
- Subduction

- Convergent/destructive plate boundary
- Transform plate boundary

REFLECTION

What revision strategies do you find most helpful when revising plate tectonics and plate boundaries? How do you remember which types of tectonic plates are involved? Is there a mnemonic that you can link to this topic to help you? For example, the mnemonic '**transform**' can be used to remember what happens at that plate boundary:

- T:** Two plates slide past each other
- R:** Rubbing edges create pressure
- A:** Along fault lines, e.g. the San Andreas Fault
- N:** No volcanoes
- S:** Shallow earthquakes
- F:** Friction builds up between plates
- O:** Often large earthquakes
- R:** Releases energy as seismic waves
- M:** Movement is sideways

« RECALL AND CONNECT 1 «

Tectonic activity can shape coastal landscapes and influence coastal hazards. Volcanic islands, like those in the Pacific Ocean, are also a result of tectonic processes shaping coastal

environments. See if you can answer the questions below from [Topic 2, 'Coastal environments'](#).

- a** Add the correct key term into the gaps in the paragraph below. Use the words provided.

corrosion attrition hydraulic action
corrasion

Waves erode coastlines through four main processes:

_____, where waves force air into cracks in rocks;
_____, where pebbles and sand grind against cliffs;
_____, where acids in seawater dissolve rock; and
_____, where rock fragments smash against each other, becoming smaller and smoother.

- b** Match the coastal feature, a–d to its correct description, 1–4.

Coastal feature		Description	
a	spit	1	A steep rock face formed by erosion at the coast.
b	stack	2	A curved, sandy landform extending into the sea, often with a hooked end.
c	headland	3	A resistant section of rock that juts out into the sea.
d	cliff	4	An isolated pillar of rock left behind after the erosion of an arch.

- c** What are two advantages and two disadvantages of hard engineering?
- d** Name two soft engineering management techniques used along coastlines.
- e** What are some of the threats to mangroves?